

U.S. Mortgage Calculator

Personal Financial Scenario Report

United States · USD

fincalcsmart.com/us-mortgage-calculator

EXECUTIVE SUMMARY

Manageable Score

\$342,000

Mortgage Amount

\$2,218.21

Monthly Payment
30-yr term

\$456,554

Total Interest
133% of principal

\$798,554

Total Cost

PRINCIPAL VS. TOTAL INTEREST (OF TOTAL MORTGAGE COST)



Total projected: \$798,554

Principal: \$342,000 (42.8%) Total Interest: \$456,554 (57.2%)

AI-ASSISTED MORTGAGE SUMMARY

Your 30-year mortgage on a \$380,000 property carries a loan of \$342,000. At 6.75%, total interest over the loan term is \$456,554 — 133% of the mortgage principal. Mortgage Health Score: 48/100 (Manageable).

Your Monthly (12×/year) payment of \$2,218.21 covers principal and interest. Total estimated monthly housing costs are \$2,835.71 (incl. property tax, home insurance, PMI).

PMI adds \$142.5/mo to your payment. Score based on down payment, PMI, and rate. Add your annual income to include DTI analysis. A 200-basis-point rate increase to 8.75% would raise your monthly payment by \$472.31.

INPUTS & ASSUMPTIONS

Home Price	\$380,000
Down Payment	\$38,000 (10.0%)
Annual Interest Rate	6.75%
Loan Term	30 years
Payment Frequency	Monthly (12×/year)
Annual Property Tax	\$4,200
Annual Home Insurance	\$1,500

MORTGAGE BREAKDOWN

Mortgage Amount	\$342,000
PMI	\$142.50/month · drops at 78% LTV (~yr 9)
Base Monthly P&I Payment	\$2,218.21/month
Monthly Payment	\$2,218.21
Total Monthly Housing Cost	\$2,835.71/month
Total Interest Paid	\$456,554
Total Mortgage Cost	\$798,554
Balance After Year 5	\$321,055
Balance at Midpoint (Year 15)	\$250,670
Mortgage Health Score	48/100 (Manageable)

KEY DRIVERS / POSSIBLE ADJUSTMENTS

- PMI adds \$142.5/mo to your payment. Score based on down payment, PMI, and rate. Add your annual income to include DTI analysis.
- PMI of \$142.50/month applies because your down payment is below 20%. PMI cancels automatically when your balance reaches 78% LTV (~year 9). Reaching 20% equity sooner allows a cancellation request under the Homeowners Protection Act.
- Prepayment opportunity: adding \$100/month toward principal saves an estimated \$66,610 in interest and may shorten your loan by approximately 3 years.

METHODOLOGY

What this calculator models:

- Uses standard US monthly compounding — divides the quoted annual rate by 12 to derive the monthly rate.
- Calculates the regular mortgage payment using the standard present-value annuity formula, then converts to bi-weekly or accelerated bi-weekly using the approved frequency multipliers.
- Applies PMI when down payment is below 20%, calculated monthly as (loan amount × annual rate) ÷ 12, and cancelled automatically when the outstanding balance reaches 78% LTV per the Homeowners Protection Act.
- Builds a year-by-year amortization schedule with extra monthly payments applied directly to principal.

What this calculator does not model:

- Closing costs, origination fees, discount points, title insurance, or other transaction costs.
- Adjustable-rate mortgages (ARMs) — this report models the full loan term at the quoted fixed rate.

EDUCATIONAL DISCLAIMER

This report is for illustrative and educational purposes only. Mortgage payment, interest, PMI, and Mortgage Health Score calculations are estimates and may differ from amounts quoted by a lender. US mortgage interest uses standard monthly compounding; actual payment amounts depend on your lender's specific calculation methods. PMI rates and cancellation timelines are estimates — actual removal is subject to lender verification of equity per the Homeowners Protection Act. DTI ratio guidelines (28% front-end, 36% back-end) are based on conventional Fannie Mae standards and may differ by loan program, lender, and credit profile. Results do not constitute financial, mortgage, lending, tax, or legal advice. Consult a licensed mortgage professional or financial advisor before making any mortgage or property purchase decisions.